InsightVLM: Automated Statistical Reasoning from Data Visualizations

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Abstract

Charts carry most quantitative findings to their readers, yet chart-understanding datasets supervise models to answer questions, describe figures or reconstruct code, never to state what the data shows and be checked against it. This thesis introduces InsightVLM-20K: 20,000 charts in 26 chart types, each labelled with one of 20 statistical findings (such as bimodality, heteroscedasticity, a level shift, or a non-significant group difference) that a named statistical test has confirmed on exactly the plotted data. 6,083 charts come from 654 real public tables in source-disjoint splits, so generalisation to real data is measured directly.

A fine-tuned CNN ensemble names the finding of a real test chart with 0.867 accuracy (95% CI 0.846–0.886) and of a synthetic chart with 0.999, against 0.468 for the best zero-shot vision–language model and 0.288 for a rule that uses no pixels. Accuracy falls to 0.716 when the same real data is drawn by a plotting library never seen in training; training on charts drawn by several libraries is the most effective remedy we found. A QLoRA-tuned generative model names findings 80% of the time but cites correct numbers only 13.8% of the time. A Grad-CAM pointing game shows the classifier's attention falls on the verified evidence beyond chance for three of four finding families.

We also report two errors found and corrected along the way: label-dependent marks in the renderer that had inflated time-series accuracy to 0.96 from a true 0.31–0.41, and an ensemble selected on the test set. The dataset, the evaluation protocol, the trained models and an interactive tester are released.

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Chapter 1

Introduction

1.1 Background

Quantitative findings travel as pictures. A reader meets a distribution, a correlation or a trend as a chart long before meeting the numbers behind it. The properties that most often invalidate a naive reading — bimodality that makes a mean meaningless, heteroscedasticity that invalidates a standard error, a group difference that is statistically significant yet practically negligible — are present in the figure but absent from its caption.

Recent Vision–Language Models (VLMs) are able to understand charts, answer questions, generate descriptions, and reconstruct underlying data. However, most existing approaches are designed for a predefined task and a user-specified query, rather than for automatically discovering the important statistical patterns in a visualization.

1.2 Problem Statement

Existing chart VLMs answer questions a human already thought to ask, describe what a figure looks like, or reconstruct the code that drew it. None of these objectives requires the model to decide what is worth reporting. Consequently users must still manually study a chart to determine its trends, relationships, distributions and anomalies.

Three specific deficiencies follow, and they organise the rest of this thesis.

(P1) No corpus carries a verified statistical finding as ground truth. Supervision in prior work is a question and an answer, a caption, or plotting code. Recent work does verify factual claims about charts [2, 7], but it does so post hoc, on captions a model has already produced, and at the level of values, labels and simple trends. No corpus pairs each chart with a statistically confirmed finding that a model could be trained toward. Without such a label there is nothing to optimise for statistical correctness, and a model may reconstruct a histogram's data table perfectly while never observing that the distribution is bimodal.

(P2) Analysis is not question answering. Autonomous analysis requires choosing what to report. A model answering a supplied question never makes that choice, so question-driven benchmarks cannot measure the capability we care about.

(P3) Claims are ungrounded. No chart corpus ties a stated finding to the specific pixels that support it. The question "was the model right for the right reason?" therefore cannot be asked, let alone answered. Geirhos et al. [5] show why this matters: without such a check, a model may reach high accuracy through features that carry no statistical evidence at all.

Who this affects. Analysts and domain scientists who read charts under time pressure and need the non-obvious property flagged; and anyone consuming automated reporting built on current chart VLMs, where a fluent but statistically wrong summary is indistinguishable from a correct one.

1.3 Scope

In scope. Twenty statistical finding classes across four analysis families (distribution shape, bivariate relationship, temporal behaviour, group comparison); single-label classification and generation from a static chart image; a balanced, statistically verified benchmark with underlying data tables, supporting statistics and pixel-level evidence; charts rendered with matplotlib/seaborn at fixed resolution.

Out of scope. Compound findings (charts exhibiting two findings at once were excluded rather than multi-labelled); causal claims; cross-renderer generalisation to Excel, Plotly, D3 or hand-drawn figures; interactive, 3-D, geographic and multi-panel dashboards; chart-type classification as an end in itself.

1.4 Motivation

Two observations motivate the work.

First, statistical correctness is not currently measurable in this field. A chart VLM can produce a fluent paragraph containing fabricated numbers and score well on every existing benchmark, because those benchmarks grade answers rather than claims. Our own experiments make this concrete: a fine-tuned VLM names the correct finding in 79% of cases while only 13.9% of the numbers it cites appear in the ground truth (Chapter 3).

Second, numeracy failures have been documented only in text. Several authors show that language models struggle with statistical operations [1, 13, 17, 21], but every such evaluation supplies the numbers directly. Whether the weakness persists when a number must first be read off a rendered chart is untested.

Third, synthetic chart corpora are now standard, and their generalisation is rarely measured. Large-scale synthetic generation has become the dominant way to scale chart datasets, but without a held-out real subset it is impossible to say how much reported performance is a generator artefact.

Table 1.1: Traceability. Every objective serves a research question and addresses a deficiency from Section 1.2; no objective exists without both.

Objective	Serves	Addresses	Measured by
O1	RQ1	P1	Label re-verification rate; generator→class fidelity
O2	RQ1	P2	Class accuracy; macro-F1
O3	RQ2	P1	Numeric fidelity vs. deterministic oracle
O4	RQ3	P3	Evidence Pointing Accuracy vs. random control
O5	RQ4	P1	Real-only probe vs. synthetic accuracy
O6	RQ5	P1, P2	Chart-type prior; oracle ceiling; 11-test shortcut audit

1.5 Research Questions

RQ1. Can a vision–language model be trained to automatically identify statistical patterns, including trends, correlations, distribution characteristics and anomalies, directly from a chart image without a supplied question?

RQ2. How does the statistical correctness of VLM-generated insights compare with a deterministic rule-based statistical pipeline applied to the same underlying data?

RQ3. To what extent do insights generated by InsightVLM remain evidence-grounded, meaning that each insight can be traced to a specific, verifiable statistical claim and its supporting chart evidence?

RQ4. How well does a model trained primarily on synthetic chart data generalise to real-world charts, as measured on a held-out real-only evaluation set?

RQ5. What evaluation protocol correctly measures statistical-pattern detection performance on chart images, given that chart type alone is structurally predictive of the finding class, and how robust is the underlying dataset to non-statistical shortcuts?

1.6 Objectives

O1. Build the multimodal chart dataset and statistical ground truth.

O2. Train and evaluate InsightVLM for autonomous statistical-pattern detection.

O3. Build the deterministic statistical baseline and compare correctness.

O4. Develop the evidence-grounding evaluation.

O5. Evaluate generalisation on real-world charts.

O6. Define a corrected evaluation protocol (chart-type-prior floor, data-oracle ceiling) and audit the dataset for non-statistical shortcuts (caption text, chart-type bias, real/synthetic provenance leakage).

1.7 Contribution Statement

1. A purpose-built multimodal chart dataset pairing chart images with underlying tabular data and statistically verified findings, providing a controlled foundation for evaluating autonomous statistical pattern discovery.

2. InsightVLM, a vision–language approach designed to automatically identify and report statistical patterns and insights from chart images without requiring a predefined user question.

3. A statistical-correctness evaluation framework that compares model-generated insights against a deterministic rule-based statistical pipeline operating on the same underlying data.

4. An evidence-grounding evaluation protocol for determining whether generated insights can be traced to specific, verifiable statistical claims and supporting chart evidence.

5. A generalisation evaluation measuring the performance of a model trained primarily on synthetic charts when evaluated on a held-out set of real-world charts.

6. A corrected evaluation protocol establishing a chart-type-prior floor and a data-oracle ceiling for scoring statistical pattern detection, together with a shortcut and bias audit verifying that the dataset cannot be solved via non-statistical cues.

1.8 Thesis Structure

Chapter 2 reviews the chart-understanding literature in seven groups and identifies the gap this work targets. Chapter 3 describes the methodology: dataset construction and its verification gate, the models, the evaluation protocol, and the risks that could invalidate the work. Chapter 4 reports the results, beginning with a correction to earlier drafts (a rendering leak and a test-set selection), and answers each research question. Chapter 5 discusses what the results mean, the lessons of the leak, threats to validity and limitations. Chapter 6 concludes and sets out future work. Appendix A describes the interactive tester and Appendix B lists the artefacts and how to regenerate every result.

Chapter 2

Literature Review

This chapter reviews 23 papers grouped by what their supervision signal actually is, rather than by model architecture. That axis is chosen deliberately: the central claim of this thesis concerns what a dataset teaches a model to do, not how the model is built.

2.1 Review Method

Papers were selected to cover (i) the benchmarks that define chart understanding, (ii) the largest and most recent chart corpora, (iii) the numeracy literature that establishes where language models fail at statistics, (iv) the two works that verify factual claims about charts, and (v) the methodological works that justify our auditing practice. Each paper was read against a nine-question extraction protocol covering problem, data, inputs, method, baselines, evaluation, results, limitations and reproducibility. Seven groups emerged.

2.2 Group 1: Chart Understanding and Recognition

Several authors approach chart understanding as a structure-recovery problem. Lee et al. [12] pretrain on screenshot parsing with variable-resolution inputs; Liu et al. [16] add mathematical reasoning and chart derendering to that pretraining, reaching 90.2 relaxed accuracy on ChartQA; Masry et al. [19] introduce a chart-grounded pretraining corpus and reach 88.56. Wang et al. [23] then show that these gains are fragile on realistic charts drawn from scientific papers.

The shared limitation is that all four target recognition. They recover what a chart shows — values, axes, structure — and answer questions about it. None proposes a statistical characterisation of the data on its own initiative.

2.3 Group 2: Chart Question Answering

Several authors established chart QA as a task by generating questions over figures: Kahou et al. [10] and Kafle et al. [9] from templates over synthetic charts, Methani et al. [20] over scientific plots with a hybrid template and crowdsourced pipeline, and Masry et al.

[18] over real charts with human-authored questions.

The limitation here is structural rather than incidental. The question exists before the model sees the figure. A model is therefore never required to determine what is notable about a chart, and the answer is a value or a comparison rather than a statement about the data's statistical character.

2.4 Group 3: Vision–Language Models and Efficient Adaptation

Our generative model builds on general-purpose multimodal architectures. Liu et al. [14] introduced visual instruction tuning and Liu et al. [15] improved its baselines; Wang et al. [24] provide the dynamic-resolution backbone we fine-tune; Dettmers et al. [3] provide the 4-bit adaptation method that makes this feasible on a single consumer GPU. These are used as tools rather than compared against as competitors.

2.5 Group 4: Statistical and Numerical Reasoning

A distinct literature measures whether language models can do statistics at all. Anon./2025 authors [1] benchmark statistical competence directly; Li et al. [13] and Mahendra et al. [17] expose systematic numeracy gaps; Various [21] attribute part of the failure to number tokenisation and propose single-token number embeddings.

Several authors converge on the same finding: models struggle with standard deviation, multiplication, exponentiation and normalisation robustness. Crucially for this thesis, every one of these evaluations is conducted in text. None asks whether the weakness persists, worsens, or changes character when the number must first be read off a rendered chart.

2.6 Group 5: Chart Datasets and Synthetic Generation

The most recent corpora scale through code-guided synthesis. Kondic et al. [11] generate chart-image/code pairs at scale; Zhao et al. [25] target chart-to-code generation with a 160k corpus; Various [22] scale explanation data by pairing chart questions with generated justifications.

In all three the supervision signal is fidelity of reconstruction or explanation. None carries a label asserting that a statistical statement about the chart's data is true. Reconstruction and analysis are different skills: a model may emit correct plotting code — correct chart type, correct data, correct styling — while holding no representation of what the data means.

2.7 Group 6: Chart Caption Factuality and Verification

Two works verify factual claims about charts, and they are the closest prior work to this thesis.

Table 2.1: Gaps identified across the reviewed literature.

	Gap	Groups exhibiting it
G1	Extractive/QA paradigm, not discovery: a question must always be supplied	G1, G2, and also G6 (both verify or correct an already-generated caption)
G2	Verification exists for caption facts, not for statistical claims	G6 addresses per-sentence entailment and simple value/ratio/trend claims only
G3	Numeracy weaknesses shown in text only, never grounded in a chart	G4
G4	Synthetic pipelines optimise reconstruction, not statistical meaning	G5
G5	No chart-specific shortcut or bias audit protocol	G7

Huang et al. [7] establish a typology of chart-caption factual errors, release the CHOCOLATE dataset of human-annotated captions, train CHARTVE as a visual entailment model for factuality scoring, and propose C2TFEC, a two-stage table-extraction and correction framework. Author(s) per TechRxiv listing [2] extract quantitative claims from a generated caption and cross-check them against the underlying chart data, reporting a substantial reduction in numerical mismatches.

These papers refute the simplest version of our motivating claim. It would be wrong to assert that verification of chart claims is an open problem: it is not. What remains unaddressed is narrower and is stated precisely in Section 2.10.

2.8 Group 7: Shortcut Learning and Survey Evidence

Geirhos et al. [5] characterise shortcut learning — features that predict a label without containing the relevant evidence — and Huang et al. [8] survey automatic chart understanding in the foundation-model era. Neither operationalises a concrete, quantitative shortcut-audit protocol for chart models. This directly motivates the eleven-test audit in Chapter 3: a balanced chart dataset can be solved by chart type, caption text or rendering style unless those channels are explicitly measured.

2.9 Synthesis: Five Gaps

2.10 The Gap This Thesis Targets

We target the narrowed form of G2, combined with G1: autonomous, unprompted discovery of statistical findings from a chart image, verified against the chart's underlying data.

This is deliberately narrower than "verification of chart claims", which Section 2.7 shows is already addressed. Three distinctions separate this thesis from Huang et al. [7] and Author(s) per TechRxiv listing [2], and all three must hold for the contribution to

stand.

(D1) Generation-first, not correction-first. Both prior works begin from a caption that already exists and check or repair it. A finding that no captioner proposed is invisible to them. InsightVLM proposes findings autonomously, so the question of what is worth reporting is part of the task rather than an input to it.

(D2) Relational and statistical claims, not per-sentence factual entailment. CHARTVE scores whether a sentence is entailed by a chart; the TechRxiv pipeline checks values, ratios and simple trends. Neither asks whether a correlation is real and how strong, whether a distribution is genuinely skewed rather than heavy-tailed, or whether a group difference is statistically meaningful rather than noise. These require a named statistical test, not an entailment score.

(D3) Verified labels in the dataset, not verification applied to output. This distinction is the reason a dataset contribution survives. Prior work contributes a verification method operating on model output; CHOCOLATE annotates errors in generated captions. Neither pairs each chart with a statistically confirmed finding as ground truth. Without such labels a model cannot be trained toward statistical correctness — only audited after the fact.

Motivating evidence from our own pipeline. This narrowing is not speculative. Our preliminary analysis (Chapter 3) finds that a fine-tuned VLM names the correct finding in 79% of cases while only 13.9% of the numeric values it cites appear in the ground truth. That is the same failure mode Huang et al. [7] and Author(s) per TechRxiv listing [2] address for basic caption facts — but measured here, for the first time, on autonomously discovered statistical findings.

2.11 Positioning

Relative to the closest prior work: Kondic et al. [11] and Zhao et al. [25] supervise reconstruction; Various [22] supervises explanation of a supplied answer; Huang et al. [7] and Author(s) per TechRxiv listing [2] supervise correction of an existing caption. This thesis supervises discovery and justification, with every training label backed by a named statistical test and every claim tied to the pixels supporting it.

Chapter 3

Methodology

This chapter describes what was built and how it was evaluated: the dataset and its verification gate (Section 3.2), how charts are rendered and what that rendering is allowed to show (Section 3.3), the models (Section 3.5) and the evaluation protocol (Section 3.4). It closes with the risks that could invalidate the work and how each was handled.

3.1 Overview of the Pipeline

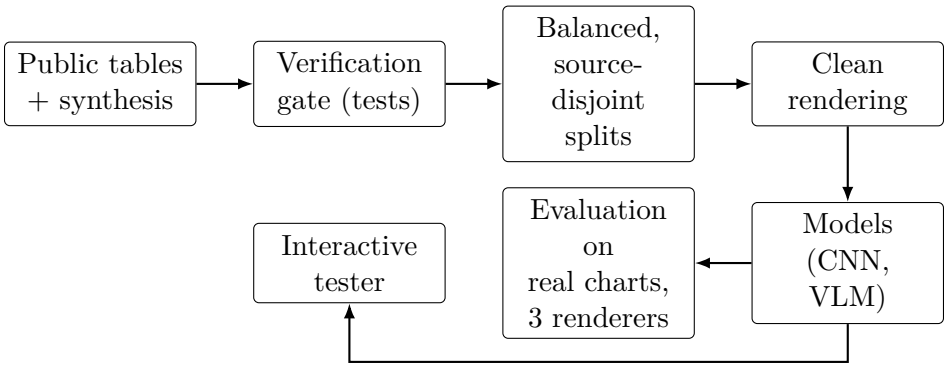


Figure 3.1: Pipeline: tables are admitted only through the verification gate, split by source, rendered without label-dependent marks, used to train models, and evaluated on real charts drawn by three plotting libraries.

3.2 Data: InsightVLM-20K

3.2.1 Task Definition

Input: a chart image. Output: the statistical finding the chart shows (one of 20 classes). Every sample additionally carries the data table, the statistics that support the finding, a reasoning chain and an evidence-grounded insight, so the same corpus supports table reconstruction, generation and grounding.

3.2.2 Label Space

Twenty finding classes span four analysis families. Each class was admitted only if it is (i) verifiable by a standard named statistical test, (ii) mutually exclusive from every other class by explicit guard conditions, and (iii) visible in a rendered chart.

Table 3.1: Finding families and their classes.

Family	n	Classes
Distribution	6	normal, right-skew, left-skew, bimodal, heavy-tail, uniform
Relationship	6	strong positive, strong negative, none, non-linear monotonic, non-linear non-monotonic, heteroscedastic
Temporal	5	increasing, decreasing, seasonal, change-point, anomaly spike
Group	3	significant difference, no difference, dominant category

3.2.3 The Verification Gate

This is the methodological core of the thesis and the mechanism by which Gap G1 (Section 2.10) is closed. A label is never asserted; it is tested. Three rules govern construction.

Rule 1 — subsample before verifying. Data is reduced to its final plotted size before the statistical test runs. Verifying a parent dataframe and then plotting a subsample would produce ground truth describing data the image does not show.

Rule 2 — exactly one class must match. A candidate satisfying two classes' guards is discarded as ambiguous rather than multi-labelled. This keeps the primary label single-valued and the confusion matrix interpretable, at the cost of excluding compound findings (see Section 3.6).

Rule 3 — rejection is free; wrong labels are not. Every guard drops a sample rather than mislabelling it. More candidates can always be mined; a poisoned label cannot be undone.

Why robust statistics are mandatory. A naive implementation of these tests fails in ways we measured, and three guards exist specifically to prevent it:

- Moment skewness is not tail-robust. Over 40 draws of a symmetric Student-$t(4)$ sample, the moment estimator alone mislabelled 12.5% as skewed. Requiring agreement with Bowley (quartile) skewness eliminated this.
- Hartigan's dip test is calibrated against the uniform, so a genuinely flat distribution trips $p < 0.05$ roughly 5% of the time. The bimodal guard therefore additionally requires a KS test against the uniform to reject.

- A monotone ramp defeats changepoint detection, producing a high-SNR "changepoint". The guard therefore requires a two-level step model to beat a linear fit and an abruptness ratio > 0.55 (measured: true step 1.15, smooth rise-then-fall 0.18, pure ramp 0.14).

Generator-to-class fidelity over 10 seeds $\times$ 20 classes is 95.5%; every remaining miss is a conservative rejection, never a wrong label.

3.2.4 Two Construction Tracks

A balanced corpus cannot be built from real data alone. Public repositories hold thousands of right-skewed columns and almost no clean inverted-U relationships: mining 719 real tables yielded exactly ten. The dataset is therefore a declared hybrid, and we state the mix rather than obscuring it.

Track R (real-seeded) mines every plottable view from 719 public tables (OpenML, World Bank, Our World in Data, vega-datasets, seaborn-data, UCI). Track S (controlled synthesis) generates data whose property holds by construction, from a domain vocabulary spanning health, economics, environment, engineering, education and retail. Both tracks pass the identical gate.

Provenance is balanced within each class rather than real-first; real-first would leave some classes almost entirely real and others almost entirely synthetic, letting a model key on provenance instead of the statistical pattern.

3.2.5 Splits and the Supplementary Real Pool

Splits are 14,000/2,000/4,000, exactly 700/100/200 per class, and source-disjoint for real data: a hash of the source-table identifier assigns every chart derived from one table to a single split. The real subsets hold 4,338 training, 553 validation and 1,192 test charts; real class frequencies are uneven because public data are.

Balance caps real data at 500 per class, so the gate admitted more real charts than the balanced splits use. We keep the surplus as a supplementary real training pool: candidates whose source table the same hash assigns to train, with exact duplicates of any dataset sample removed and at most 400 per class. The build asserts that no validation or test source contributes. The pool adds 2,545 real training charts in nine classes, raising real training data from 4,338 to 6,883.

3.3 Rendering

Charts are drawn with matplotlib and seaborn on a fixed 768 $\times$ 576 canvas across 26 chart types and 8 visual styles. Each sample's style is chosen from a CRC32 hash of its identifier, so rendering is reproducible across processes.[1] Grounding boxes marking the evidence (outlier band, modal bin, change point, anomalous points, dominant wedge) are emitted in image pixel coordinates. Tight bounding-box cropping is deliberately disabled:

[1]The first version used Python's built-in hash(), which is salted per process, so styles could not be reproduced. This was corrected in version 2.

Table 3.2: Dataset summary and integrity audit.

Samples	20
Classes	20, exactly 1,000
Chart types	
Provenance	6,083 real (654 source tables) / 13,917 synth
Supplementary real training pool	2,545 (train-split sources o
Underlying data points	8,487
Label re-verification from stored data	100.
Leaked sources across splits	0 of
Duplicate data fingerprints	
Real test charts with near-duplicate data in training (Sec. 4.6)	85 of 1
Grounding boxes outside canvas	

it resizes the canvas after the axes transform is read, silently invalidating every box. Dates stored as epoch timestamps are plotted as calendar dates.

3.3.1 No Label-Dependent Marks (Version 2)

The first release drew annotations on time-series charts that depended on the label: a fitted trend line only on increasing and decreasing series, a vertical line at the change point only on level-shift series, and circles only around anomalies. The seasonal class carried no mark, so the absence of a mark was itself informative. Such marks let a model identify the class without reading the data. Version 2 draws no label-dependent marks; the evidence survives only as grounding boxes in the metadata. No other family drew label-dependent marks (regression lines on scatter plots are drawn independently of the label). Chapter 4 measures how strongly models had exploited the marks. All results in this thesis use version 2.

3.3.2 Cross-Renderer Test Sets

All training charts come from one plotting library, so a model could learn matplotlib's appearance rather than the statistics. The 1,192 real test tables were therefore re-drawn by two more libraries from the same stored data: Plotly, with five templates (excluding the one that imitates seaborn), and Vega-Lite, with nine themes (excluding those that match a matplotlib training style). Chart types map to each library's nearest native form; where none exists the substitute is a form the taxonomy lists as valid for every class that chart type can carry, so no label becomes unreadable.

3.3.3 Training-Only Renderer Augmentation

Every training and supplementary chart was also rendered twice more, for training only: by Plotly, and by matplotlib with randomised fonts, font sizes, line and marker sizes, colours, background tints, gridlines, spines and title placement. The data and chart type are unchanged, so the label is too. No validation or test chart is rendered this way and no Vega-Lite render is used in training, so Vega-Lite remains a renderer that no model has seen, and the Plotly test set becomes a seen-renderer test for augmented models.

3.4 Evaluation Protocol

This section addresses RQ5 and objective O6.

3.4.1 Why the Conventional Baseline Is Wrong Here

With 20 balanced classes, chance is 5%. But chart type is associated with the finding by construction — a pie chart can only carry a composition finding, a QQ plot only a distribution finding. A model consulting no pixels at all, predicting the most common class for each chart type, reaches **0.288**. Reporting against 5% would credit a model with roughly 24 points it did not earn. We therefore use the chart-type prior as the floor.

3.4.2 A Data Oracle Gives the Ceiling

Running the verification battery on each sample's stored data table — seeing the numbers, not the picture — yields **0.9995**. This is the deterministic rule-based pipeline of RQ2/O3, and a vision model is interesting only within $[0.288,\ 0.9995]$.

3.4.3 Real Charts Are the Headline

Because the corpus is majority synthetic, mixed-test accuracy cannot distinguish learning the statistics from learning the generator. The headline metric is accuracy on the 1,192 real test charts (RQ4/O5), reported for the matplotlib render and for the Plotly and Vega-Lite re-renders. Every accuracy carries a 95% bootstrap confidence interval (4,000 resamples), and differences between systems scored on the same charts are tested with exact McNemar tests. Every choice of model, checkpoint, ensemble member and hyperparameter is made on the validation split; the test split is scored once.

3.4.4 Shortcut and Bias Audit

Eleven failure modes are measured against thresholds fixed in advance (Table 3.3). All pass on version 2.

Provenance detectability matters most: with 69.6% of samples synthetic, the obvious objection is that a model learns to recognise generated charts. A classifier on low-level pixel statistics separates real from synthetic at AUC = 0.551 — close to chance. The audit examines metadata and pixel statistics, however, not what a trained model attends to; it did not detect the version-1 marks (Section 3.3.1), which were found by inspecting the renderer. We therefore added a direct ablation (Section 4.1).

3.5 Models

3.5.1 Discriminative Models (O2)

ImageNet-pretrained ResNet-101, ResNet-50 and DenseNet-121 are fine-tuned at 384×384 for up to 12 epochs with AdamW (learning rate 3×10^{-4}, weight decay 0.05), a one-cycle

Table 3.3: Shortcut and bias audit on version 2; all eleven tests pass.

Test	Value	Threshold
Class balance (imbalance ratio)	1.000	$= 1.000$
Split balance (max s.d.)	0.0000	< 0.01
Chart type $\rightarrow$ label (pixel-free prior)	0.2883	reported as floor
Render style $\rightarrow$ label (Cramér's V)	0.0000	< 0.05
Provenance detectability (AUC)	0.551	< 0.75
Axis-text shortcut (macro-F1)	0.3688	< 0.55
Pixel-statistic shortcut (macro-F1)	0.4038	< 0.55
Source concentration (top share)	17.2%	$< 35\%$
Length confound (within family)	2.04	< 4.0
Subject-domain balance	22.8%	$< 45\%$
Category-vocabulary spread	13.0%	$< 35\%$

schedule, label smoothing 0.1 and mixed precision on two T4 GPUs. Two choices target real-chart accuracy directly:

- Real-data oversampling. A weighted sampler draws about half of every batch from real charts (31% of the balanced training split is real).
- Real-validation checkpointing. The checkpoint kept is the epoch with the highest accuracy on the 553 real validation charts, with mixed validation accuracy as the tie-break. Mixed accuracy saturates near 0.95 and says little.

Models are trained in three steps, each an ablation of the next: on the balanced split; with the supplementary real pool added; and with renderer augmentation added (each epoch shows each training chart as one of its three renders, uniformly).

Augmentation constraint. Horizontal and vertical flips are excluded. Mirroring a scatter converts a strong positive correlation into a strong negative one, and mirroring a line chart converts an increasing trend into a decreasing one — both separate classes here, so the default augmentation would silently corrupt roughly 30% of labels. Only label-preserving transforms (mild crop and scale, colour jitter, blur, small erasures) are used.

Ensemble. The deployed model averages the class probabilities of the full-recipe members (supplementary pool and renderer augmentation) with equal weights. Probabilities are averaged rather than logits, whose scales differ across architectures. A label-shift correction towards real-data class frequencies, estimated from training data only, is available with its strength tuned on validation (it is not used: the tuned strength is zero). Greedy ensemble selection on validation and the average of all trained models are reported as alternatives.

3.5.2 Zero-Shot Vision–Language Models

To test whether current general-purpose VLMs can already do the task, Qwen2.5-VL-3B, Qwen2.5-VL-7B and Qwen3-VL-8B receive each real test chart with all 20 class names

and one-line definitions and must answer with one class name (greedy decoding, no task training). Answers are parsed by exact label match; over 99.9% parse.

3.5.3 Generative Model: InsightVLM (O2, O3)

Qwen2-VL-2B adapted with QLoRA ($r = 16$, 4-bit NF4, 18.5M trainable parameters, 0.83% of the model), trained one epoch on 4,000 samples. The supervision target is the reasoning chain plus the insight, so the model learns to produce evidence-grounded prose rather than a class identifier. Numeric fidelity is the share of numbers the model cites that match the ground-truth statistics to two decimals. It was first trained on version-1 renders and re-trained on version 2 with the same settings once the leak was found; Chapter 4 reports the version-2 model.

3.5.4 Evidence-Grounding Evaluation (O4)

For each real test chart that carries at least one evidence box, the Grad-CAM map of each deployed member for its predicted class is computed at the last convolutional block, upsampled, and its peak mapped back to image pixels. A hit is a peak inside any evidence box. The control is a uniformly random point inside the plot area, whose expected hit rate equals the evidence boxes' share of the plot area, so large evidence regions cannot inflate the result.

3.5.5 Interactive Tester

A Gradio web application (Appendix A) loads the deployed ensemble and runs uploaded charts through exactly the pipeline that was scored. It reports the finding, its confidence, how many members agree, and the statistical test that defines the class, and can draw random held-out test charts to show predictions against the verified ground truth.

3.6 What Could Go Wrong

R1. Synthetic majority.
: 69.6% of samples are synthetic; balance forced this. Mitigation: provenance balanced within each class, identical gate for both tracks, real charts reported as the headline, and a supplementary real pool for training. Residual risk: the real subset may not represent real charts broadly.

R2. Single-label design.
: Compound findings were excluded, not handled, so the model will not learn to report a trend and an anomaly together. Mitigation: stated as a limitation; multi-label extension is future work.

R3. Difficulty axis does not track difficulty.
: Accuracy does not follow the evidence-margin tertiles (Spearman 0.034). Extreme margins produce degenerate charts — a residual spread ratio of 1537 renders as a single vertical smear. Mitigation: reported as a finding about chart benchmarks rather than presented as a working difficulty axis.

R4. Renderer monoculture.
Training charts come from one plotting library. Mitigation: Plotly and Vega-Lite test sets measure the effect, and training-only renderer augmentation reduces it; Vega-Lite is held out entirely.

R5. Thresholds are conventions.
$|r| > 0.7$ as "strong" and $\varepsilon^2 > 0.14$ as "large" are choices. Mitigation: all cuts are recorded explicitly so results can be re-derived under alternatives.

R6. Template recitation by the VLM.
The training target embeds each class description verbatim, so the model could score well by memorising a phrase. Mitigation: the real/synthetic gap is the diagnostic — observed 0.683 vs 0.844, comparable in magnitude to the CNNs, indicating chart reading rather than recitation.

R7. Compute constraints.
Training ran on free Kaggle T4 GPUs with a weekly quota; the generative result used 4,000 samples, one epoch and 2B parameters at 4-bit. Mitigation: reported as a compute-constrained floor, not a ceiling.

R8. Rendering leaks.
A renderer can reveal the label without the audit noticing. This happened (Section 3.3.1). Mitigation: version 2 removes every label-dependent mark, and a with/without ablation quantifies the effect.

R9. Selection on the test set.
An earlier ensemble was chosen by ranking combinations on test accuracy, which inflates the reported number. Mitigation: all selection now uses validation only (Section 3.4).

3.7 Reproducibility

The corpus rebuilds end to end with run_all.py; 07_validate.py (integrity) and 10_bias_audit.py (shortcuts) audit it. The clean renders, the supplementary pool, the Plotly and Vega-Lite renders and the augmentation renders are produced by 06_render.py, 12_extra_real.py, 11_render_plotly.py and 13_render_vega.py. Training notebooks are generated by train/make_split_notebooks.py, and every reported number comes from final_ensemble.py, stats_report.py, grounding_eval.py and ablate_marks.py. All thresholds live in a single module so results can be re-derived under alternative conventions. Documentation follows the datasheet standard of Gebru et al. [4].

Chapter 4

Results

This chapter reports every result on the held-out test split, scored once, with selection done on validation only (Section 3.4). It begins with a correction to results reported in earlier drafts, because every later number depends on it, and then answers the five research questions in turn.

4.1 A Correction: Two Problems in the Earlier Results

Earlier drafts of this work (Chapter 3, version 1) reported an ensemble reaching 0.865 on real test charts. That number had two problems.

Selection on the test set. The ensemble was chosen by ranking every combination of trained models on test accuracy. Selecting the same models on validation instead gives 0.845. The difference is small, but a number chosen on the test set is not a test result.

A rendering leak. Version-1 time-series charts carried marks that depended on the label (Section 3.3.1). To measure how much the models relied on them, the same 1,000 test time series were rendered twice with identical styles, once with the marks and once without, and scored with two ResNets trained on version 1 (Table 4.1, Figure 4.1).

Table 4.1: Accuracy of version-1 models on test time series rendered with and without the label-dependent marks. Only the marks differ between the two renders.

	ResNet-101		ResNet-50	
Test charts	with marks	without	with marks	without
Real time series ($n = 305$)	0.957	0.311	0.944	0.407
Synthetic time series ($n = 695$)	1.000	0.801	1.000	0.911
level shift	1.000	0.355	1.000	0.705
increasing	0.975	0.570	0.930	0.570
seasonal (never marked)	0.970	0.970	0.990	0.990

Without the marks, accuracy on real time series fell from 0.957 to 0.311: the models had largely learned to read the annotation rather than the data. The seasonal class, the only temporal class that never carried a mark, was unaffected, which isolates the cause.

The zero-shot vision–language models of Section 4.2, which never saw version-1 training data, changed by only 1–2.5 points on all real charts and 4–10 points on real time series. This is shortcut learning [6] of the plainest kind, and the shortcut audit (Table 3.3) did not detect it, because the audit examines metadata and pixel statistics rather than what a trained model relies on.

Every result from here on uses version 2, trains new models on it, and selects on validation only.

4.2 RQ1: Can a Model Identify the Finding from the Chart Alone?

Table 4.2 gives the main results and Figure 4.2 the comparison. The deployed ensemble (the equal-weight average of the two full-recipe models) reaches 0.867 on the 1,192 real test charts (95% CI $[0.846, 0.886]$) and 0.999 on synthetic ones, 0.959 over all 4,000 test charts. Macro-F1 on real charts is 0.776, lower than accuracy because the rarest real classes are also the hardest. Retraining both members with two further random seeds, and nothing else changed, gives real-chart accuracies of 0.867, 0.863 and 0.858 (mean 0.863 ± 0.004, s.d. over 3 seeds); Plotly 0.857 ± 0.006, Vega-Lite 0.713 ± 0.012 and synthetic 0.999 ± 0.000 (train/seed_variance.py). The deployed run is the best of the three on matplotlib charts but within one standard deviation of the mean, so training variance is small next to the test-set interval.

Table 4.2: Accuracy on the 1,192 real test charts drawn by three plotting libraries, and on synthetic test charts. Brackets: 95% bootstrap CI. †Fixed 400-chart synthetic sample.

	Real test charts			
System	matplotlib	Plotly	Vega-Lite	Synthetic
Chart-type prior (no pixels)	0.288 (all test charts)			
Qwen2.5-VL-3B, zero-shot	0.306 [0.281, 0.332]	–	–	0.303†
Qwen2.5-VL-7B, zero-shot	0.411 [0.383, 0.440]	–	–	0.498†
Qwen3-VL-8B, zero-shot	0.468 [0.439, 0.496]	–	–	0.648†
ResNet-101, balanced split	0.841	0.747	0.607	0.997
ResNet-50, balanced split	0.828	0.722	0.608	0.998
ResNet-101, + supplementary pool	0.840	0.735	0.602	0.999
ResNet-50, + supplementary pool	0.845	0.760	0.599	0.998
DenseNet-121, + supplementary pool	0.861	0.771	0.638	0.997
ResNet-101, + pool + renderer aug.	0.859	0.854	0.650	0.998
DenseNet-121, + pool + renderer aug.	0.855	0.852	0.712	0.998
Ensemble (two full-recipe models)	0.867 [0.846, 0.886]	0.857 [0.836, 0.877]	0.716 [0.690, 0.741]	0.999
Statistical oracle on the data table	0.9995 (all test charts)			

Zero-shot vision–language models cannot do this task reliably. The best, Qwen3-VL-8B, reaches 0.468; the smallest, Qwen2.5-VL-3B, is barely above the 0.288 floor that uses no pixels at all. The ensemble is right on 511 charts where Qwen3-VL-8B is wrong and wrong on only 36 where it is right (exact McNemar $p < 10^{-100}$; the same holds for the other two models). Naming a statistical finding from a chart is therefore not a capability current general-purpose VLMs already have.

How the result was built. Each step in Table 4.2 changes one thing. Adding the supplementary real pool helped ResNet-50 (+1.7 points) but not ResNet-101 (−0.1); DenseNet-121 trained on the pool was the strongest single model on matplotlib charts (0.861). Renderer augmentation raised ResNet-101 from 0.840 to 0.859 and, as the next section shows, mattered far more on other renderers. Choosing the ensemble differently changes little: averaging all seven trained models gives 0.862, greedy selection on validation 0.865.

Where the model fails. Figures 4.3 and 4.4 break the result down. Errors stay almost entirely within a family, and the family with the most errors is distribution shape (0.763 on real charts, against 0.886–0.930 for the others). The commonest confusions are between classes that differ in degree rather than kind: bimodal read as right-skewed (19 of 100), heavy-tailed read as right-skewed (13 of 26) and significant group differences read as none (12 of 100). Accuracy is lowest where real examples are rare: heavy tails (0.385, $n = 26$), seasonality (0.462, $n = 13$), left skew (0.474, $n = 19$), and the inverted-U relationship, of which only two real test charts exist. Per-class results for the rarest real classes are indicative only: with $n = 13$ to 26 real test charts, a class's 95% interval spans roughly ± 19 to ± 27 points.

Answer to RQ1. Yes, for most findings. A fine-tuned vision model names the statistical finding of a real chart correctly 86.7% of the time without being asked a question, far above both the pixel-free floor and current zero-shot VLMs. It remains unreliable for subtle distribution shapes and for classes with little real data.

4.3 RQ2: Model Insights versus a Deterministic Statistical Pipeline

The deterministic pipeline is the verification battery run on the chart's stored data table. It recovers the finding for 0.9995 of test charts, and the two exceptions lie within 1% of a guard threshold and fall to no class rather than to a wrong one. Its numbers are exact by construction.

The generative model (Qwen2-VL-2B with QLoRA, Section 3.5) was evaluated on a 300-chart test subset after training on version-2 renders. It named the correct finding for 0.787 of charts (95% CI $[0.740, 0.833]$; 0.659 on the 82 real charts, 0.835 on synthetic), but only 13.9% of the numbers it cited matched the ground-truth statistics. It has learned the vocabulary and structure of statistical reasoning while largely inventing the supporting numbers. The same model trained on version-1 renders scored 0.800, with 1.000 on time series against 0.947 on version 2 and identical results on the other families: the leak inflated its time-series accuracy, as it did for the CNNs, while numeric fidelity was unchanged (0.138 against 0.139).

Answer to RQ2. Far below. The rule-based pipeline is essentially always right and exact; the generative model is right about the finding four times in five but cites correct numbers about one time in seven. Fluent statistical language is much easier to produce than true statistical language, and only a dataset with verified statistics can tell them apart.

4.4 RQ3: Is the Model Looking at the Evidence?

We tested whether the deployed models attend to the evidence that licenses a label, using the pointing game of Section 3.5: for each of the 1,023 real test charts that carry at least one evidence box, a hit is a Grad-CAM peak inside an evidence box, and the control is a random point in the plot area (Table 4.3).

Table 4.3: Pointing game on 1,023 real test charts. Gain is the pointing rate minus the random-point control, with a 95% bootstrap CI.

Charts	Pointing	Random control	Gain [95% CI]
Temporal ($n = 305$)	0.649	0.495	+0.154 [+0.139, +0.168]
Group ($n = 258$)	0.599	0.489	+0.110 [+0.064, +0.153]
Relationship ($n = 173$)	0.642	0.569	+0.073 [+0.029, +0.115]
Distribution ($n = 287$)	0.253	0.254	−0.002 [−0.035, +0.033]
ResNet-101 member	0.543	0.439	+0.105 [+0.081, +0.130]
DenseNet-121 member	0.504	0.439	+0.066 [+0.042, +0.089]
All	0.524	0.439	+**0.085** [+0.068, +0.102]

The models point at the evidence more often than chance overall (+8.5 points), most clearly for time series (+15.4) and group comparisons (+11.0), less for relationships (+7.3), and not at all for distributions (−0.2, CI spanning zero). Pointing is slightly better when the prediction is correct (0.570 and 0.533 for the two members).

Two readings are possible for the distribution family, and the data cannot yet separate them. A distribution's shape is a property of the whole histogram or density rather than of one region, so an evidence box around the modal bin or the outlier band may simply be the wrong reference for where a model should look. Alternatively, the models may judge shape from global cues. Either way, distribution is also the family with the lowest accuracy (0.763), so weak grounding and weak accuracy coincide.

Answer to RQ3. Partly. The model's attention is tied to the verified evidence beyond chance for three of four families, but only modestly: about half of all peaks fall inside an evidence box, against 44% for a random point. The CNN classifiers are not generating insights, so this measures whether their decisions rest on the evidence, not whether a generated sentence is traceable; the generative model's traceability is limited by its 13.8% numeric fidelity (Section 4.3).

4.5 RQ4: Generalisation to Real-World Charts

Real-world generalisation has two parts: real data, and charts drawn by other tools.

Real data. On matplotlib charts the gap between synthetic and real test charts is 13.2 points (0.999 against 0.867). Real charts are harder: 44% of real samples fall in the hardest evidence-margin tertile against 29% of synthetic ones, and real data are noisier and less clean-cut than generated data.

Other plotting libraries. Trained on matplotlib alone, every model loses 8.5–10.6 points when the same real data is drawn by Plotly and 22–25 points when it is drawn by Vega-Lite (Table 4.2). The models had learned part of matplotlib's appearance. Training-only renderer augmentation (Plotly renders plus randomised matplotlib styles) closes most of the Plotly gap, as expected for a renderer seen in training, and, more tellingly, lifts accuracy on Vega-Lite, which no model has seen, from 0.60–0.64 to 0.65–0.71, without costing matplotlib accuracy. The deployed ensemble reaches 0.857 on Plotly and 0.716 on Vega-Lite.

Answer to RQ4. A model trained on a mostly synthetic corpus generalises to real data well (0.867) when the chart is drawn the way its training charts were, and reasonably (0.716) when it is drawn by a tool it has never seen. The 15-point unseen-renderer gap is the main open problem this dataset exposes, and renderer diversity in training is the most effective remedy we found.

4.6 RQ5: Evaluation Protocol and Shortcuts

The protocol of Section 3.4 changed the conclusions in three ways.

1. The floor. A rule using no pixels reaches 0.288, not 0.05. The weakest zero-shot model (0.306) is barely above it, which a 5% baseline would have hidden.
2. Real charts as the headline, three renderers. Mixed accuracy (0.959) hides a 13-point real/synthetic gap and a 15-point unseen-renderer gap, both of which are the numbers that matter for use.
3. Selection on validation only. Selecting on test inflated the earlier result by two points; selection on 553 real validation charts is itself noisy, so the deployed ensemble uses equal weights and the alternatives are reported.

The eleven-test shortcut audit passes on version 2 (Table 3.3); in particular, pixel statistics cannot tell real from synthetic charts (AUC 0.551). But the audit missed the rendering leak of Section 4.1: an audit of data and pixels does not replace testing what a trained model uses.

Near-duplicates across splits. The integrity audit removes exact duplicates and keeps source tables split-disjoint, but the same data can be published under two sources. Every real validation and test chart was therefore compared with every real chart a model can train on (training split and supplementary pool) of the same arity: candidate pairs whose data signatures correlate above 0.99 count as near-duplicates only when the data itself matches (for time series, mean and spread within 5%; for sampled data, at least half the (x, y) pairs, or 90% of the values, identical; for compositions, the same categories and shares). 85 real test charts (7.1%) and 34 validation charts are flagged. Most are republished tables, such as World Bank and Our World in Data internet-use series or two OpenML uploads of the same dataset; a few are coincidental matches of integer-valued columns. On the flagged charts the ensemble scores 0.918; without them its real-chart accuracy is 0.863 (95% CI $[0.843, 0.883]$) against 0.867 with them, so the overlap does not drive the headline result.

Answer to RQ5. Report accuracy on real charts, drawn by more than one renderer, against a pixel-free floor and a data-oracle ceiling, with confidence intervals, paired tests and validation-only selection; and test trained models directly for shortcuts, because a data audit alone did not catch the one that mattered.

4.7 Human Validation

A label check asked whether a person, looking only at the chart, agrees with the finding its label asserts. 188 real validation and test charts (up to 10 per class) were each shown with their labelled finding, mixed with 25 decoys shown with a deliberately wrong finding from the same family, in one fixed random order with anonymous identifiers. Annotators answered yes, no or cannot tell and did not see each other's answers or the statistics behind a label. The exclusion rule was fixed before data collection: an annotator who accepts more than 40% of the decoys is excluded. Two annotators completed it (one is an author of this thesis), using an offline spreadsheet version that did not record response times; each also completed the sheet a second time.

Table 4.4: Label check: two annotators, 188 real labels and 25 decoys. The interval is a 95% bootstrap over charts.

Labels confirmed (yes)	0.737 [0.676, 0.790]
Labels rejected (no)	0.168
Cannot tell	0.096
Decoys rejected / accepted	0.78 / 0.04
Agreement between annotators (raw / Fleiss' κ)	0.867 / 0.68
Agreement with own second pass	0.84, 0.85
Labels rejected by both annotators	25 of 188

Neither annotator met the exclusion rule. Both rejected the decoys and confirmed most labels, with substantial agreement (Table 4.4). Confirmation is not uniform across classes: it is 0.95–1.00 for the clear-cut findings (normal, uniform and right-skewed distributions, trends, dominant category, no group difference, inverted-U), but 0.25 for bimodal and heteroscedastic, and 0.40 for monotonic non-linear and anomaly-spike charts. The labels in those classes are statistically correct (re-running the tests reproduces them), so a rejection means that the finding is true of the data but not evident to a person from the chart at this size: a two-component mixture with modest separation, or a residual spread that widens by the required ratio, is easy to test and hard to see. Bimodal charts are also the models' commonest error (Section 4.2). We read this as a property of the task, not as label noise: the dataset labels what the data shows, and some of what the data shows is subtle. With two annotators, one of them an author, the study is a validity check rather than a full annotation campaign, and per-class rates rest on 10–20 judgements each. A human classification baseline (choosing one of the 20 findings) was designed but not completed.

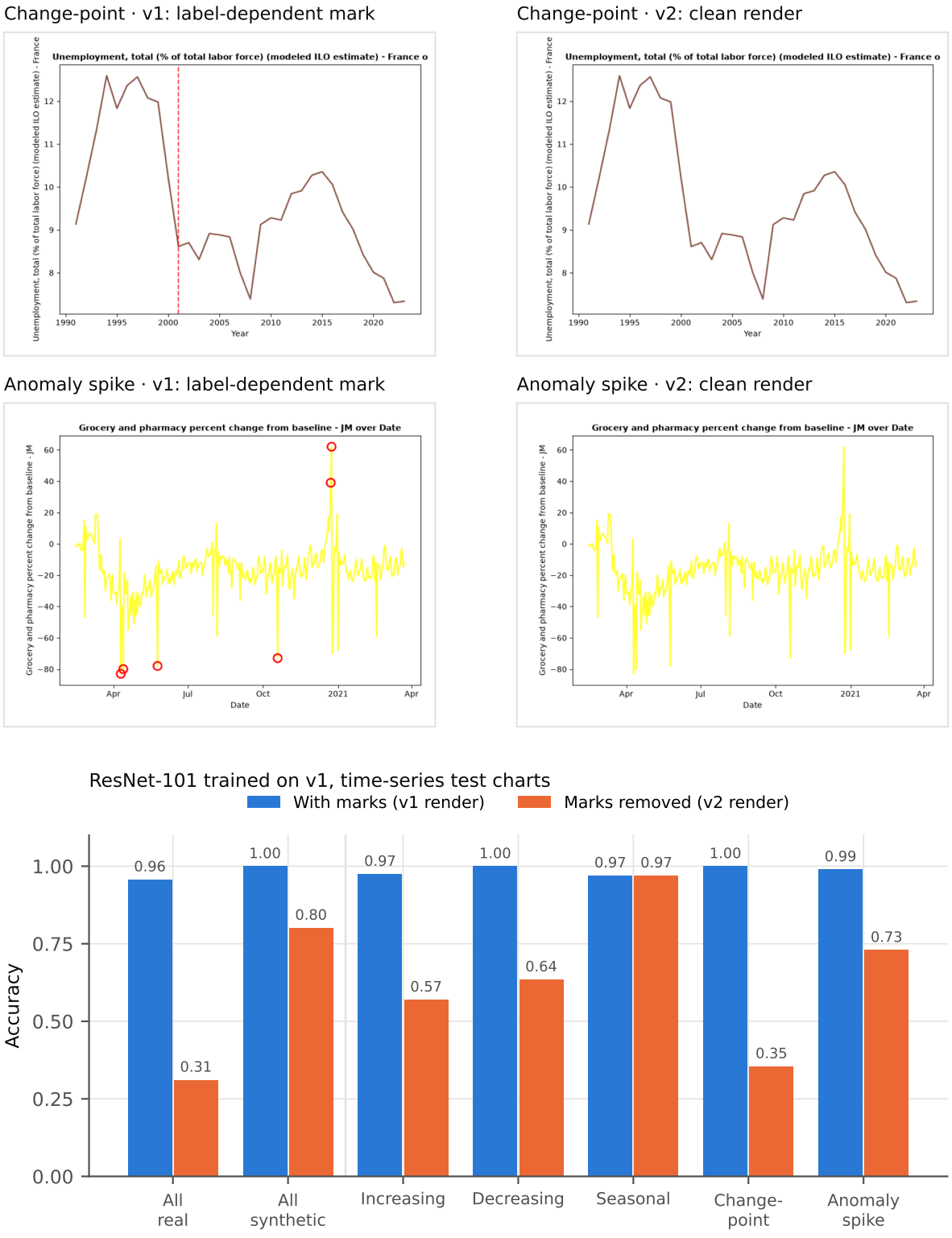


Figure 4.1: Top: the same real data rendered with a version-1 label-dependent mark and in the released version-2 form. Bottom: accuracy of a version-1 ResNet-101 with and without the marks.

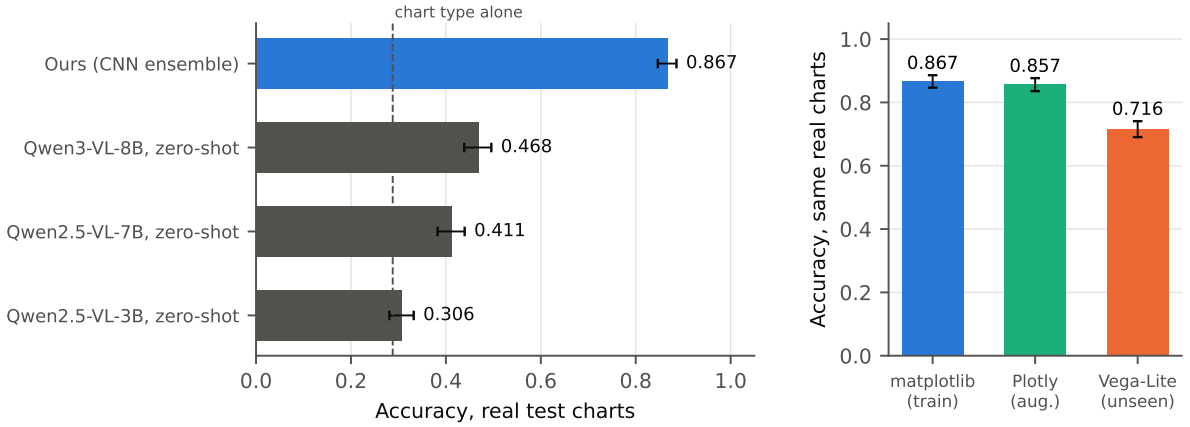


Figure 4.2: Left: accuracy on the real test charts (matplotlib renders) with 95% bootstrap CIs; the dashed line is the chart-type prior. Right: the deployed ensemble on the same real charts drawn by the training renderer, by Plotly (used only as training augmentation) and by Vega-Lite (never seen).

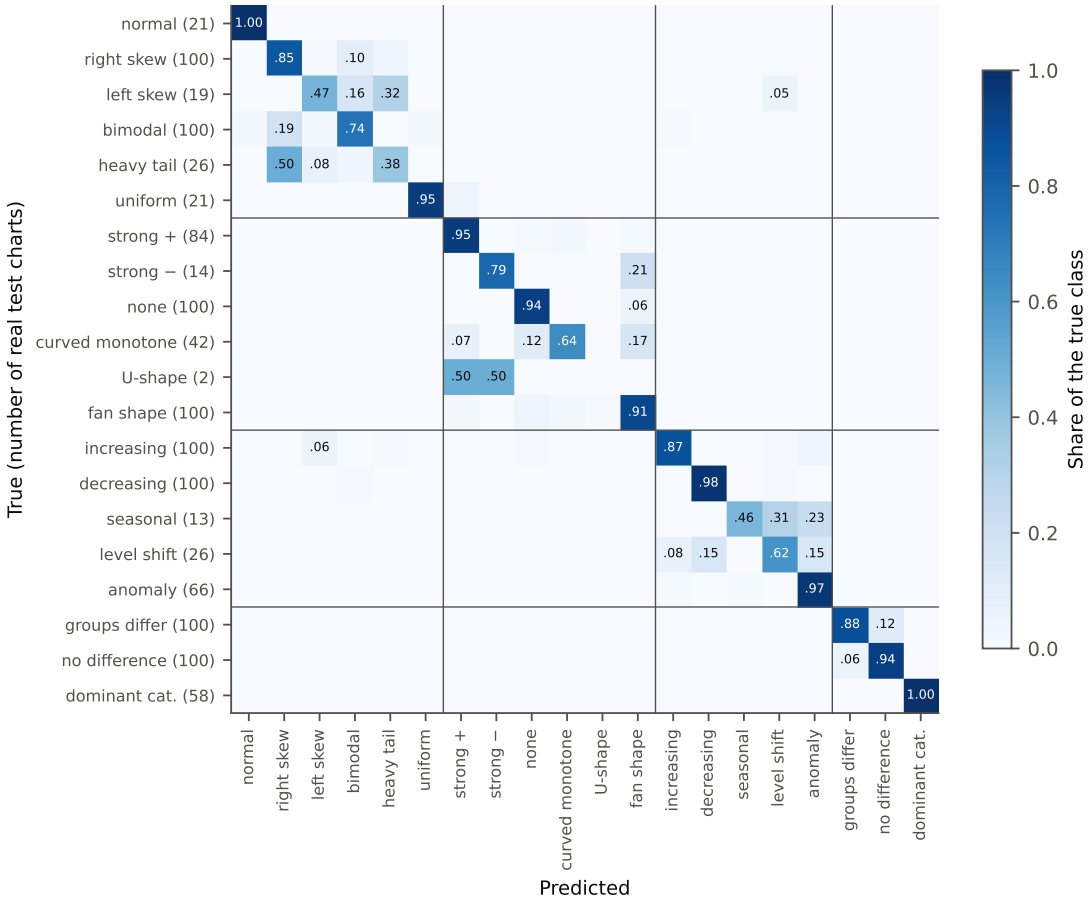


Figure 4.3: Confusion matrix of the deployed ensemble on the 1,192 real test charts, normalised by true class; lines separate the four families. The number of real charts per class is in brackets.

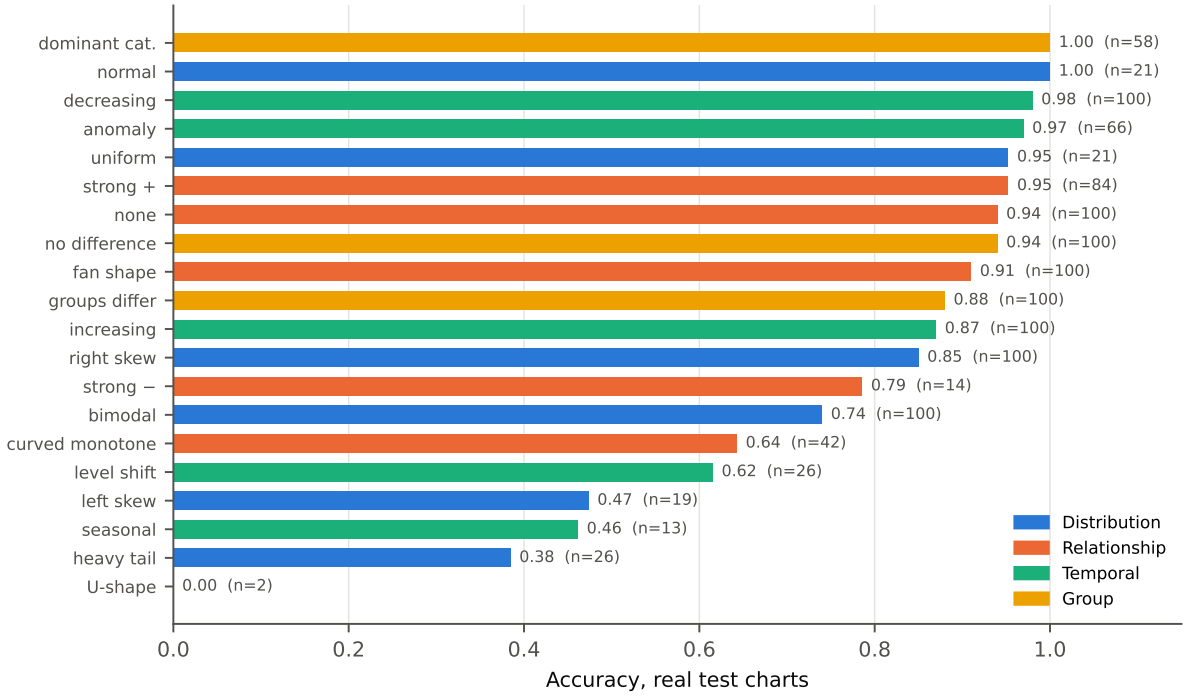


Figure 4.4: Accuracy of the deployed ensemble per class on the real test charts.

Chapter 5

Discussion

5.1 What the Results Mean

The task is learnable, and not yet solved by general models. A fine-tuned convolutional network names the statistical finding of a real chart correctly almost nine times in ten, while the strongest open vision–language model we tested manages fewer than one in two without task training. The gap is not about recognising the chart type (a rule that uses only chart type reaches 0.288): it is about judging shape, spread and change, which is exactly what the verified labels supervise. This supports the central premise of the thesis (Chapter 1): a dataset whose labels state what the data shows, rather than what a question asks, trains a capability that question-answering corpora do not.

Most remaining errors are judgements of degree. The confusions that remain (bimodal against right-skewed, heavy-tailed against right-skewed, significant against no group difference) are between classes separated by a threshold on a continuous statistic. Near a threshold, the chart itself is ambiguous: a right-skewed histogram with a shoulder can look bimodal to a person too. Some of these errors are therefore likely irreducible from pixels at this resolution; the oracle reaches 0.9995 only because it reads the numbers.

Where the training data came from mattered as much as the model. The three largest improvements in this thesis all came from the data rather than from architecture: removing the rendering leak (which made the problem honest), adding real training charts, and drawing training charts with more than one library. Architecture changes within the CNN family moved real accuracy by one to two points.

5.2 Lessons from the Rendering Leak

The leak is the most important methodological lesson of the project, and we report it in full because it could affect other synthetic chart datasets.

1. A helpful annotation can be a label. The trend line, change-point line and anomaly circles were added to make charts more informative, and they were drawn only where the finding held. Any rendering step that looks at the label is a potential leak, however natural it seems.

2. A data audit does not replace a model audit. Eleven shortcut tests on metadata and pixel statistics all passed while trained models were reading the marks. The failure was only visible by changing the input a trained model sees and measuring what happened.

3. A class that cannot carry the shortcut is a control. Seasonal series never carried a mark, and their accuracy did not move when marks were removed. That single class separated "the models read the marks" from "the renders changed".

4. High accuracy on an easy-looking class is a warning. Level-shift charts scored 1.000 with marks and 0.355 without. A perfect score on a class that should be hard deserves suspicion, not celebration.

5.3 Threats to Validity

Internal validity. All selection used the validation split, but the validation split has only 553 real charts, so choices made on it are noisy; this is why the deployed ensemble uses equal weights and why alternatives are reported. The choice to deploy the full-recipe models was made after all models were trained; the alternatives (averaging every model, 0.862; greedy selection, 0.865) bound its effect at under a point. Kaggle probability outputs were checked against a local re-computation (100% prediction agreement) before use.

Construct validity. The labels encode conventional thresholds ($|r| > 0.7$ for "strong", $\varepsilon^2 > 0.14$ for "large"). A model that scores well has learned these conventions, which are reasonable but not universal. The evidence boxes used for grounding mark where the evidence is by construction, not where a human would look.

External validity. Real test charts come from six public repositories and are drawn by our renderers, not taken from papers or reports in the wild. The Vega-Lite test set measures an unseen renderer, not unseen chart conventions (hand-drawn charts, screenshots, charts with annotations of their own). Real class frequencies are uneven, and four classes have fewer than 20 real test charts, so their accuracies have wide uncertainty.

Statistical conclusion validity. Every headline accuracy carries a bootstrap confidence interval, and system comparisons use paired exact McNemar tests on the same charts. Per-class accuracies for rare classes are reported with their n and should not be over-read. Per-class results for the rarest real classes are indicative only: with $n = 13$ to 26 real test charts, a class's 95% interval spans roughly ± 19 to ± 27 points. The confidence intervals reflect test-set sampling; training variance across random seeds is reported separately (Section 4.2).

5.4 Limitations

1. Mostly synthetic. 69.6% of the balanced corpus is synthetic; only ten clean inverted-U relationships exist in 719 real tables.

2. Single label. Charts showing two findings were excluded, so models will not report a trend and an anomaly together.

3. A small generative model. The generative result comes from a 2B-parameter model trained for one epoch on 4,000 samples and scored on 300 charts (82 real), so its real-chart accuracy has a wide interval.

4. Small human study. Two annotators, one an author, checked 188 labels (Section 4.7); they confirmed 73.7%, but only 25–40% for four classes whose findings are statistically true yet hard to see. No human classification baseline was completed.

5. Compute. All training used free T4 GPUs under a weekly quota, which limited the number of models, epochs and seeds. Only the deployed recipe was repeated over three seeds (real-chart s.d. 0.004); other configurations are single runs.

Chapter 6

Conclusion and Future Work

6.1 Summary

This thesis set out to make charts carry verified statistical findings, and to test whether a model can learn to state those findings from the image alone. We built InsightVLM-20K, 20,000 charts in 26 chart types whose labels are issued only when a named statistical test confirms them on exactly the plotted data, with 6,083 charts from 654 real public tables held in source-disjoint splits and a supplementary pool of 2,545 further real training charts.

RQ1. A fine-tuned CNN ensemble identifies the finding of a real chart with 0.867 accuracy (95% CI $[0.846, 0.886]$) and of a synthetic chart with 0.999, without a supplied question. Zero-shot vision–language models reach at most 0.468.

RQ2. A deterministic pipeline on the data table is right 99.95% of the time and exact. A QLoRA-tuned generative VLM names the finding 78.7% of the time but cites correct numbers only 13.9% of the time.

RQ3. The model's Grad-CAM peak falls inside a verified evidence box on 52.4% of real charts against 43.9% for a random point: grounded beyond chance for time series, group comparisons and relationships, but not for distributions.

RQ4. Real data costs 13 points relative to synthetic data. A plotting library never seen in training costs a further 15 points (0.716), reduced but not removed by training on charts drawn by more than one library.

RQ5. Accuracy should be reported on real charts, across more than one renderer, against a pixel-free floor (0.288) and a data-oracle ceiling (0.9995), with selection on validation only; and trained models must be tested for shortcuts directly, because a data audit missed the one that mattered.

Along the way we found and removed a rendering leak that had inflated accuracy on real time-series charts to 0.94–0.96 from a true 0.31–0.41, and corrected an earlier result that had been selected on the test set. The final numbers are lower than the earliest drafts reported, and they are the ones we can defend.

6.2 Deliverables

- InsightVLM-20K version 2, with datasheet, bias audit, the supplementary real pool, and Plotly and Vega-Lite re-renders of the real test charts.
- Trained models and the deployed ensemble.
- An interactive tester (Appendix A) that runs uploaded charts through the evaluated pipeline.
- The full build, training and evaluation code, from which every number in this thesis is regenerated.

6.3 Future Work

1. Extend the human study to more independent annotators, add the human classification baseline on 200 charts, and test whether labelling the subtle classes by visual salience as well as by statistics changes what models learn.
2. Close the unseen-renderer gap with more rendering libraries in training (Excel-style, ggplot2, D3) and with charts taken from real documents.
3. Scale the generative model to a larger backbone and the full training set, and optimise it for numeric fidelity, not only class accuracy.
4. Multi-label findings, so a chart can show a trend and an anomaly at once.
5. More real data for rare classes (heavy tails, seasonality, left skew, inverted-U relationships), which are also the weakest classes.
6. Seed variance for every configuration, not only the deployed recipe, so ablation differences can be judged against training variability.

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Appendix A

The InsightVLM Chart Tester

The tester is a local web application (app.py, built with Gradio) for trying the model on any chart. It loads the deployed ensemble from results/final_ensemble.json and runs every input through exactly the pipeline that produced the reported numbers: resize to 384×384, each member's softmax, equal-weight average, and the (zero-strength) label-shift correction.

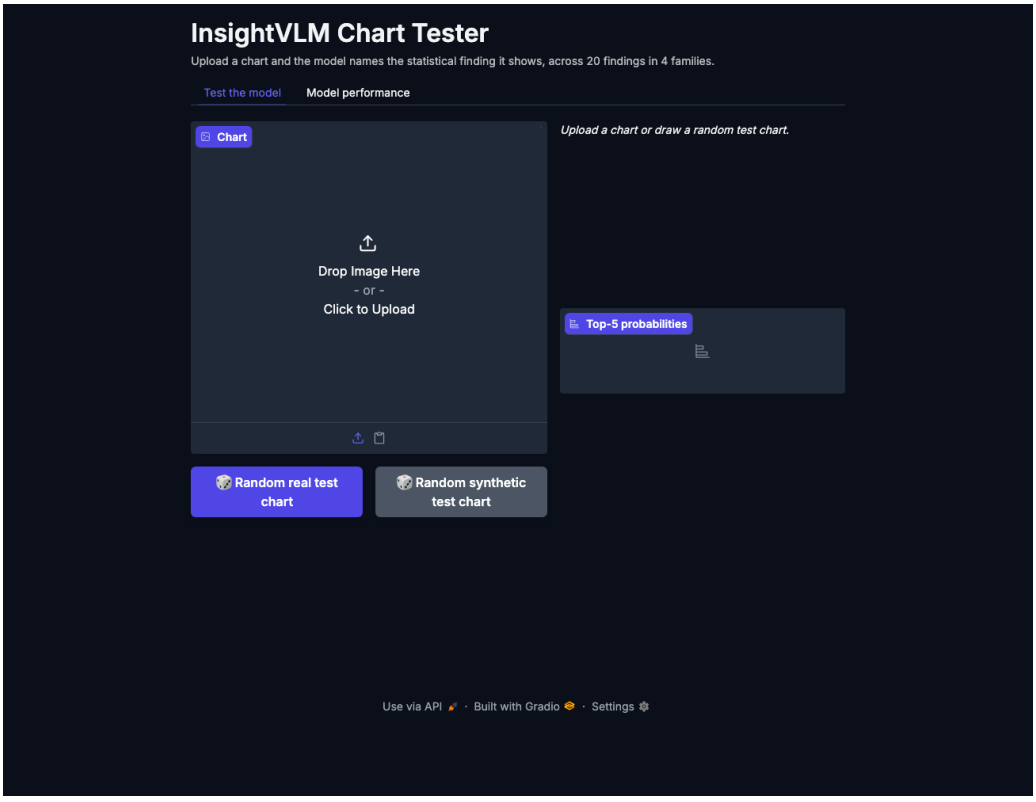


Figure A.1: The tester's main tab. A chart can be uploaded, pasted from the clipboard, or drawn at random from the held-out real or synthetic test charts.

Running it.

```
pip install torch torchvision gradio pandas pyarrow pillow
python app.py              # then open http://127.0.0.1:7860
```

What it shows. For an uploaded chart:

- the predicted finding in plain language, its confidence, and its family;
- how many ensemble members agree;
- the definition of the finding and the statistical test that licenses it;
- the five most probable findings;
- a warning when confidence is below 50%, since the chart may show a mix of patterns or one outside the 20 classes.

The random real test chart and random synthetic test chart buttons draw a held-out test chart, predict it, and show the verified ground truth, its source, chart type, difficulty and verified insight beside the prediction, marked correct or wrong. The Model performance tab lists the ensemble's test accuracy on real and synthetic charts against the zero-shot baselines and the no-pixel floor.

Limits. The tester classifies into the 20 known findings only. Charts from tools unlike the training renderers are less reliable (Section 4.5), and compound findings are reported as a single class.

Appendix B

Artefacts and Reproduction

Every number in this thesis is produced by a script in the project repository (Table B.1). Training ran on Kaggle (two T4 GPUs) from notebooks generated by train/make_split_notebooks.py; everything else runs on a laptop CPU or Apple-silicon GPU.

Table B.1: Where each result comes from.

Result	Command or file
Dataset build and audit	python run_all.py; build/07_validate.py; build/10_bias_audit.py
Clean (version-2) renders	build/06_render.py –clean
Supplementary real pool	build/12_extra_real.py, then 06_render.py –records cache/records_extra.jsonl
Plotly / Vega-Lite test renders	build/11_render_plotly.py; build/13_render_vega.py
Augmentation renders	11_render_plotly.py –split train; 06_render.py –randstyle
Model training	notebooks nb5_real, nb5x_*, nb7_*
Zero-shot VLM baselines	notebooks nb6_zeroshot, nb6b_zeroshot_ts
Leak ablation (Table 4.1)	train/ablate_marks.py
Probabilities per model and renderer	train/cache_probs.py [–view plotly|vega]
Deployed ensemble	train/final_ensemble.py → results/final_ensemble.json
Main table, CIs, McNemar	train/stats_report.py → results/stats_report.json
Grounding (Table 4.3)	train/grounding_eval.py → results/grounding.json
Figures	paper/make_figures.py
Human label check (Table 4.4)	train/human_study_eval.py (answers and key kept private)

Data. InsightVLM-20K version 2 is on Kaggle (asifxzaman/insightvlm-20k-v2), with the supplementary pool (insightvlm-extra-real) and the augmentation renders (insightvlm-aug-train). The datasheet (DATASHEET.md) and audit report (BIAS_AUDIT.md) ship with the data.